

Meteorological Studies carried out by IMD Team during XII Antarctic Scientific Expedition at Maitri

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Abstract

During XII Antarctic Expedition, regular synoptic observations were recorded every three hours and six-hourly observations were transmitted to New Delhi over telex for real-time use. Continuous recording of atmospheric pressure, temperature, wind speed, wind direction, surface ozone, global solar radiation were made during this period, in addition to atmospheric turbidity observations. A total number of 47 ozonesonde ascents, 21 radiosonde ascents, 17 low level radiosonde ascents and 9 pilot balloon ascents were taken during the year. Some significant weather occurrences pertaining to ozone-hole phenomenon during 1993 and severe blizzard have been analysed in detail and presented in this report. We also participated in Regional Ozone Soundings in Antarctica (ROSA-93) Programme from September 1-10, 1993.

Introduction

The India Meteorological Department (IMD) has been continuously participating in all the Antarctic Expeditions starting from the first expedition in 1981-82. This is an ongoing programme of recording and analysing the meteorological data, with a view to establish a suitable data base, study the changes in meteorological parameters from year to year and its implication on Indian/global weather conditions.

Scientific objectives of the meteorological programme for 12th Antarctic Expedition included study of various meteorological parameters over Antarctica in general, with particular emphasis on the following observational programme:

1. Recording the daily, seasonal and annual variations of atmospheric pressure, surface wind, surface air temperature and cloud cover.
2. Pilot balloon observations for upper-air data of the Antarctic winds.

3. Study of the atmospheric changes with reference to specific synoptic patterns.
4. Study of the radiation budget.
5. Monitoring and archival of information received through weather satellite regarding weather systems affecting the Antarctic continent.
6. Study of atmospheric turbidity in Antarctica.
7. Regular maintenance of Data Collection Platform (DCP) for transmission of hourly surface meteorological parameters on real time basis through INSAT to IMD at New Delhi.
8. Balloon borne measurements of ozone for its vertical profile over Maitri, Antarctica for the study of ozone-hole phenomena over Antarctica during spring months.
9. Slow-rising, low level radiosonde ascents for supplementing Boundary Layer Experiment of National Physical Laboratory, New Delhi.

Material and Methods

To achieve above objectives, the programme was divided into four groups: (i) Surface observations, (ii) Upper-air observations, (iii) Forecasting services and (iv) Additional services.

Surface Observational Programme

- (i) Surface observations of all weather parameters were recorded every three hours on all days and four out of eight were transmitted to India over telex on real time basis.
- (ii) Continuous recording of atmospheric pressure, temperature, wind speed, wind direction, surface ozone, global solar radiation were made during the period.
- (iii) Sunphotometer observations were taken on all clear days for analysis of the atmospheric turbidity.

Upper-air Observations

The following upper-air soundings were taken during the Expedition:

Ozonesonde ascents	47 Nos.
Radiometersonde ascents	21 Nos.

Low level radiosonde ascents	17 Nos.
Pilot balloon ascents	9 Nos.

Forecasting Services

Local weather forecasting services were provided for all outdoor scientific activities. Members were specially briefed about weather outlook when on convoys and GSI traverses, including ice-core drilling programme.

Additional Services

IMD team also contributed to the other scientific activities of the station as detailed below:

- (i) Attending to IIG work which included observational programme, data recording and transmission, auroral photography and complete maintenance of the concerned equipment.
- (ii) Helping GSI team in calibrating sensors for their glaciological studies and repairing and part maintenance of the ice-core drilling equipment.

Data related to surface weather, vertical distribution of ozone and blizzards are presented in this report. Due to shortage of time to process all the data collected, the data related to low level radiosonde, radiometersondes, solar radiation and turbidity measurements are not presented.

Besides three hourly synoptic observational data, continuous records of the weather parameters like temperature, wind speed and direction, surface ozone and global solar radiation were recorded and this data is under processing. The climatic data of 1993 for Maitri is presented in Table 1 and mean monthly variations in pressure, temperature and wind speed are presented in Figs. 1-3. Interesting weather parameters observed during the year are also presented in Tables 2-4. Two graphs relating to ozonesonde ascents data are presented in Figs. 4 and 5.

Results and Discussion

Some of the individual parameters are discussed below, highlighting their variations and ranges.

Mean Sea Level Pressure

The mean sea level pressure dips to its lowest value in August. Both mean maxima and mean minima (for the month) were lowest in August. The lowest pressure of 946.9 hPa was recorded on 4 August. Similarly, highest value of

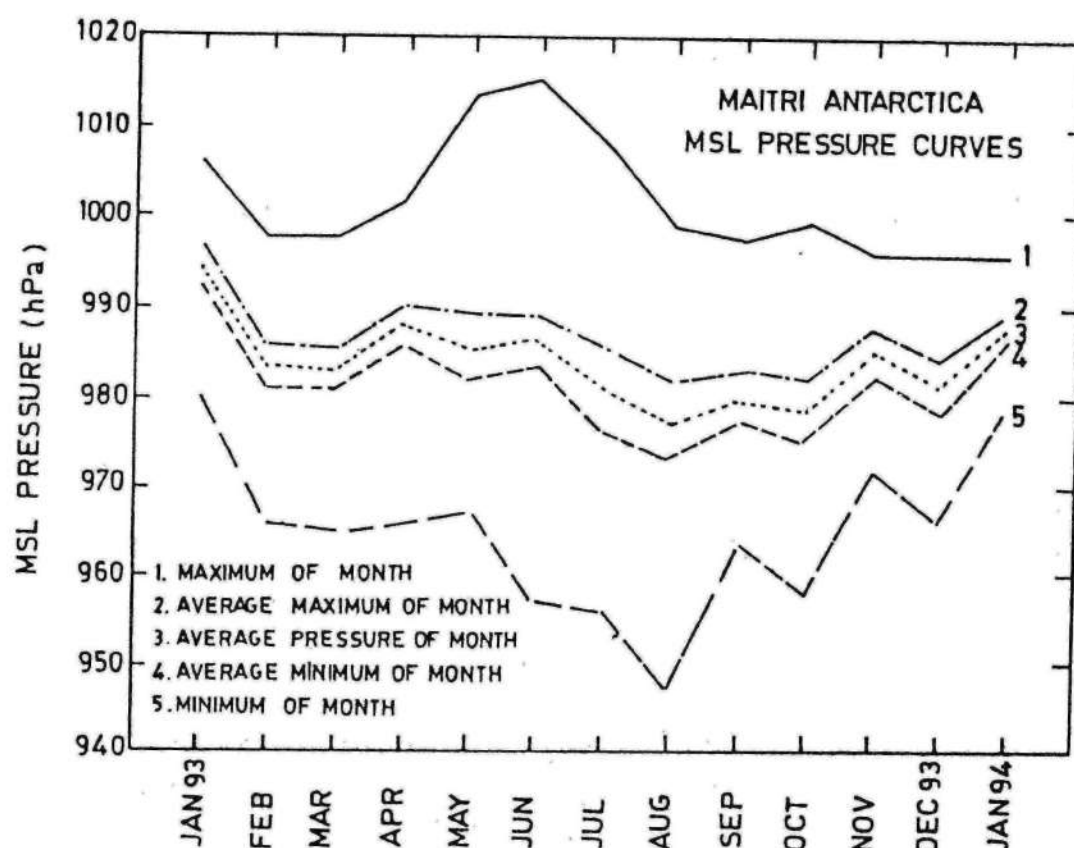


Fig. 1

mean, mean maxima and mean minima are recorded during January. However, highest maximum of 1015.1 hPa was recorded on 1 June. An approximate annual mean pressure of 970 hPa was taken as reference for monitoring day to day pressure variations.

Air Temperature

Mean monthly temperature, mean maxima and mean minima showed a gradual but rapid fall from January to July and thereafter steady rise till December. The lowest minimum temperature of -34.5°C was recorded on 28 August and a highest maximum temperature of $+08.2^{\circ}\text{C}$ was recorded on 12 January.

Surface Wind

The mean monthly wind varied from 11 to 22 knots (1 knot = 1.8 kmph) during the year 1993. September and October were the windiest months (average wind of 22 knots). Mean monthly wind remained above 15 knots

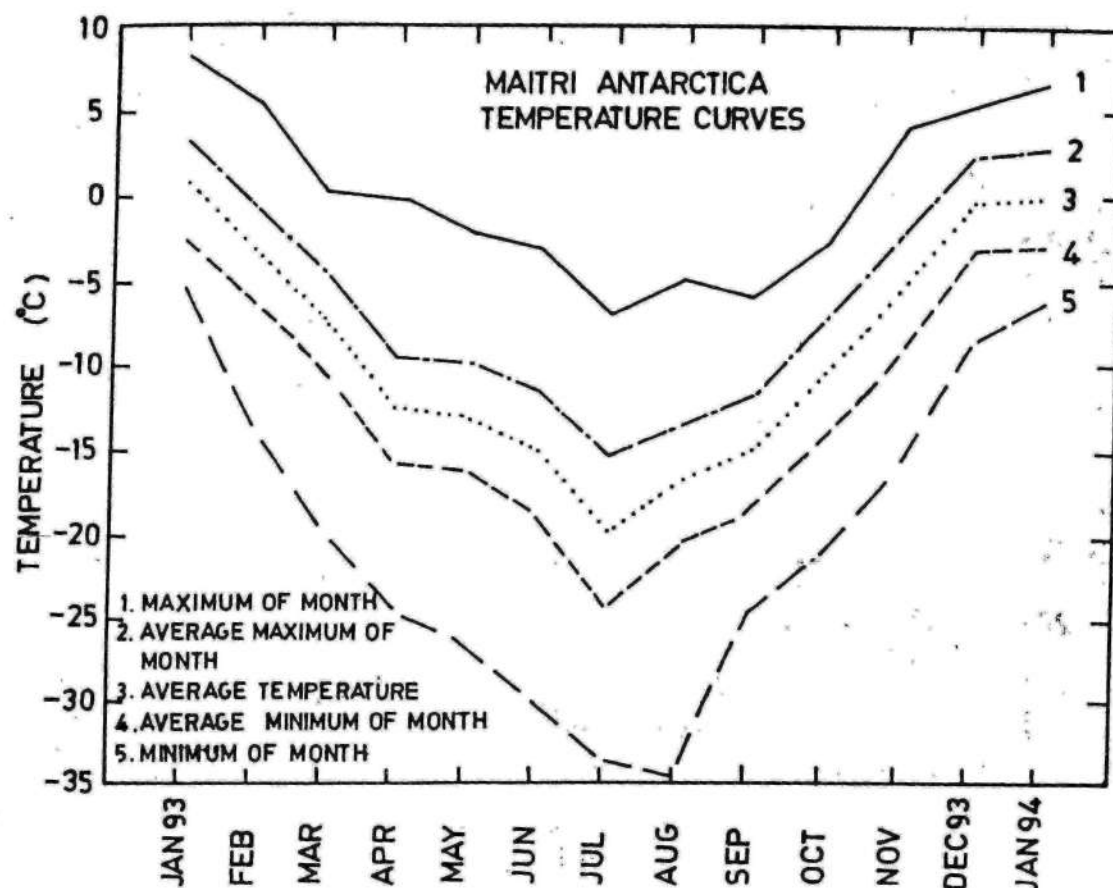


Fig. 2

throughout the year except in January (11 knots). July recorded a lower mean wind compared to June and August (Table 1).

Surface Wind Direction

Analysis of surface wind direction (Tables 2-4) showed that Maitri station gets only a few preferred wind directions. They are basically SE, S and SSE during winter and ESE and SE directions during summer. Though there are other wind directions present, their frequencies are less and correspond basically to light winds. For monitoring the correct synoptic wind a very high tower like the one (28 metre high) used by NPL scientists is required. There were a large number of calm wind days during cold winter months. Annual wind direction analysis showed that the Easterly to Southerly wind directions were most prominent at Maitri station.

Blizzards

During 1993, a great number of blizzards were experienced at Maitri, though a good number of them were short in duration and weak in intensity.

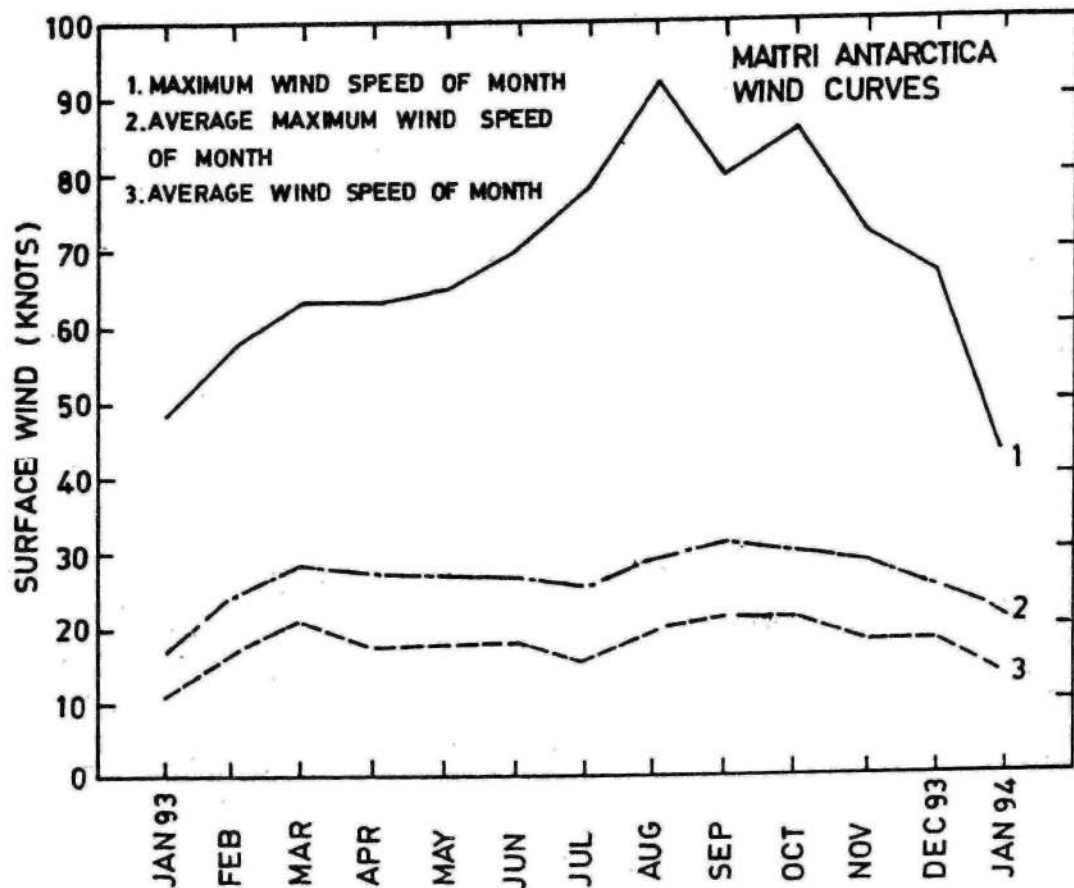


Fig. 3

For the classification of the weather system as a blizzard the following criteria was considered during the period:

- (i) Wind speed equal to or greater than 30 knots.
- (ii) Moderate or severe blowing snow present.
- (iii) Visibility at surface level less than 1000 m.

Sometimes the criteria of wind speed is relaxed if the other two are satisfied and the time of commencement and time of cessation are marked for each blizzard. A table of blizzard history (Table 2) gives an idea of the structure of each blizzard. Though blizzards occurred from April to November, a good number of them were in July and September. The longest blizzard was experienced in June, which lasted for 127 hours. The severest blizzard occurred during July-August when the maximum wind speed touched 92 knots in gust. A detailed meteorological description of this blizzard is given as a case study with relevant records.

Table 1 : Important Meteorological Statistics — Maitri-1993

Month	MSL Pressure	Avg. Temp.	Avg. Max. Temp.	Avg. Min. Temp.	Avg. Wind	Max. Avg. Wind	No. of Blzds	Longest Blzd hours	Total Blzd days	Snow fall days	No. of Aurora days
January	994.0	+00.7	+03.7	-02.7	11.3	44	0	0	0	7	0
February	983.0	-03.5	-00.9	-06.4	17.0	42	0	0	0	8	0
March	982.6	-07.5	-04.8	-10.0	21.0	48	0	0	0	4	8
April	987.5	-12.6	-09.7	-16.0	17.5	46	1	19	1	3	15
May	984.7	-13.3	-10.1	-16.4	18.0	44	3	65	7	9	4
June	986.8	-15.2	-11.6	-18.9	18.3	50	2	127	8	8	3
July	980.7	-19.8	-15.5	-24.3	15.6	60	6	47	11	13	6
August	977.1	-16.9	-13.6	-20.6	19.7	66	6	29	7	9	7
September	976.9	-14.3	-12.1	-18.6	22.0	56	4	55	12	12	6
October	978.5	-10.4	-07.4	-14.0	22.0	60	5	27	7	13	2
November	984.9	-05.5	-02.3	-09.1	18.4	55	1	46	3	3	0
December	981.3	-00.4	-02.2	-03.2	18.2	48	0	0	0	11	0

Table 2: Blizzard History at Maitri, Antarctica

Sl. No.	Time of Commencement Time/Date (U.T.C.)	Time of Cessation Time/Date (U.T.C.)	Min. M.S.L. Pressure hPa	Max. Temp. deg C	Avg. M.S.L. pressure of month hPa	Avg. Temp. of month deg C	Max. Wind Gust recorded Knots
1	1610/24-4-93	0300/25-4-93	965.3	-07.5	987.5	-12.6	62
2	0230/23-5-93	2130/23-5-93	984.8	-11.6	984.7	-13.3	40
3	1510/24-5-93	0830/27-5-93	972.1	-09.1	do	do	58
4	0830/28-5-93	0915/29-5-93	968.9	-03.7	do	do	57
5	0840/09-6-93	1600/14-6-93	956.8	-05.8	986.8	-15.2	70
6	1100/20-6-93	1600/21-6-93	959.7	-16.9	do	do	42
7	0330/05-7-93	0230/07-7-93	964.0	-09.5	980.7	-19.8	48
8	0045/09-7-93	1000/10-7-93	987.5	-08.0	do	do	50
9	1810/13-7-93	0300/14-7-93	974.4	-23.4	do	do	36
10	0200/18-7-93	1730/18-7-93	959.9	-16.2	do	do	48
11	1600/21-7-93	0930/22-7-93	957.7	-16.0	do	do	52
12	1705/31-7-93	1900/01-8-93	953.0	-10.0	977.1	-16.9	92
13	1000/02-8-93	1400/02-8-93	975.7	-10.9	do	do	50
14	1105/04-8-93	1330/04-8-93	947.4	-07.2	do	do	48
15	1215/12-8-93	1000/12-8-93	976.3	-16.4	do	do	50

Contd.

Table 2: Blizzard History at Maitri, Antarctica — Contd.

Sl. No.	Time of Commencement Time/Date (U.T.C.)	Time of Cessation Time/Date (U.T.C.)	Min. M.S.L. Pressure hPa	Max. Temp. deg C	Avg. M.S.L. pressure of month hPa	Avg. Temp. of month deg C	Max. Wind Gust recorded Knots
16	1610/14-8-93	2300/14-8-93	954.7	-08.5	do	do	53
17	0500/21-8-93	0900/22-8-93	965.0	-10.0	do	do	50
18	2030/06-9-93	0330/07-9-93	-	-	976.9	-14.3	48
19	1510/17-9-93	0400/18-9-93	975.3	-10.0	do	do	80
20	1950/21-9-93	1310/24-9-93	963.1	-11.5	do	do	76
21	2310/25-9-93	1500/27-9-93	974.0	-12.8	do	do	58
22	1115/05-10-93	1600/05-10-93	959.9	-07.5	978.5	-10.4	86
23	2300/13-10-93	0200/15-10-93	972.3	-07.0	do	do	50
24	1330/15-10-93	2300/15-10-93	-	-06.0	do	do	40
25	0200/28-10-93	1400/28-10-93	971.8	-07.2	do	do	39
26	1850/22-10-93	2200/23-10-93	974.8	-11.0	do	do	70
27	2200/05-11-93	2000/07-11-93	971.6	-04.0	984.9	-05.5	72

Table 3: Cloud Cover Frequency at Maitri, Antarctica

Month	Cloud Cover in Oktas (1/8 Sky)								Over Obscured	
	Clear	1/8	2/8	3/8	4/8	5/8	6/8	7/8	cast	
Jan 93	05	30	16	25	15	10	23	54	74	00
Feb 93	21	24	10	11	08	10	08	59	73	00
Mar 93	64	33	13	10	06	08	09	60	42	00
Apr 93	63	47	18	07	02	02	05	35	51	04
May 93	49	20	10	05	02	05	06	38	83	30
Jun 93	54	33	11	06	04	12	03	24	63	31
Jul 93	52	35	23	18	03	05	11	18	62	20
Aug 93	26	48	25	15	03	12	08	20	73	14
Sep 93	20	24	13	11	02	10	06	39	102	19
Oct 93	26	28	15	07	05	06	13	42	83	21
Nov 93	66	55	10	14	01	10	09	39	25	08
Dec 93	02	42	08	09	14	09	15	68	82	00
Percentage Occurrence										
Annual	15	14	06	05	02	04	04	17	28	05
Winter (Mar-Aug)	21	15	07	04	01	03	03	13	26	07

Contd.

Table 3: Cloud Cover Frequency at Maitri, Antarctica — Contd.

Wind Direction Frequency At Maitri, Antarctica																			
Wind Direction																			
Month	99	00	02	05	07	09	11	14	16	18	20	23	25	27	29	32	34	36	
	VRB	CLM	NNE	NE	ENE	E	ESE	SE	SSE	S	SSW	SW	WSW	W	WNW	NW	NNW	N	
Jan 93	01	26	03	08	06	42	59	57	16	07	00	02	01	00	00	00	02	15	
Feb 93	00	04	02	04	04	24	53	66	16	37	11	01	02	00	00	00	00	01	
Mar 93	00	11	01	01	01	01	09	56	42	70	30	00	01	02	01	01	00	02	
Apr 93	01	48	01	00	15	09	16	31	19	43	38	03	08	03	03	00	00	01	
May 93	00	47	01	01	14	29	32	74	14	14	08	01	02	00	02	00	00	00	
Jun 93	01	57	01	01	01	13	21	73	24	24	16	05	02	00	00	00	00	01	
Jul 93	01	56	00	00	01	12	31	53	23	40	24	02	00	00	00	00	00	00	
Aug 93	00	51	00	00	00	07	29	55	26	43	25	03	01	04	03	00	00	00	
Sep 93	00	23	01	00	00	18	66	67	20	35	08	00	00	00	00	00	00	00	
Oct 93	01	07	00	02	03	45	60	63	15	33	10	02	02	01	01	02	00	00	
Nov 93	06	19	00	05	06	31	22	53	15	50	13	01	00	02	04	06	00	00	
Dec 93	04	16	01	08	15	39	39	77	21	17	01	01	00	02	00	01	00	06	
Annual Percentage																			
	01	14	00	01	02	10	16	27	09	12	03	01	01	01	01	00	00	01	

Table 4: Data of Successful Ozonesonde Ascents Taken at Maitri, Antarctica during 1993

Sl. No.	Date of Ascent	Time of Ascent U.T.C.	Maximum Level Reached hPa	Maximum Ozone Level hPa	Maximum Partial Pr. of Ozone nano Bar
1	14-1-93	2236	104	104	143
2	23-1-93	1750	023	023	138
3	21-2-93	2140	019	064	130
4	09-3-93	1630	003	053	133
5	21-3-93	1310	009	029	117
6	27-4-93	1137	055	077	105
7	04-5-93	1248	029	059	143
8	19-5-93	1550	017	067	152
9	20-7-93	1612	013	063	206
10	19-8-93	1943	010	093	149
11	23-8-93	1833	018	045	104
12	29-8-93	1600	043	064	121
13	01-9-93	2140	057	084	142
14	03-9-93	2027	010	075	122
15	04-9-93	2049	007	110	094
16	05-9-93	2105	041	061	196
17	08-9-93	2045	029	082	095
18	10-9-93	0053	016	094	096
19	14-9-93	1553	019	330	099
20	20-9-93	1727	143	235	068
21	24-9-93	1605	073	210	064
22	7-10-93	1353	005	051	046
23	16-10-93	1726	007	018	042
24	20-10-93	1803	006	021	080
25	24-10-93	1731	020	150	042
26	09-11-93	1202	003	019	065
27	15-11-93	1320	010	018	084
28	21-11-93	1411	007	034	070
29	29-11-93	1410	005	010	143
30	17-12-93	1147	005	050	171
31	21-12-93	1521	005	037	161
32	27-12-93	1751	008	053	127
33	08-01-94	2225	005	049	148

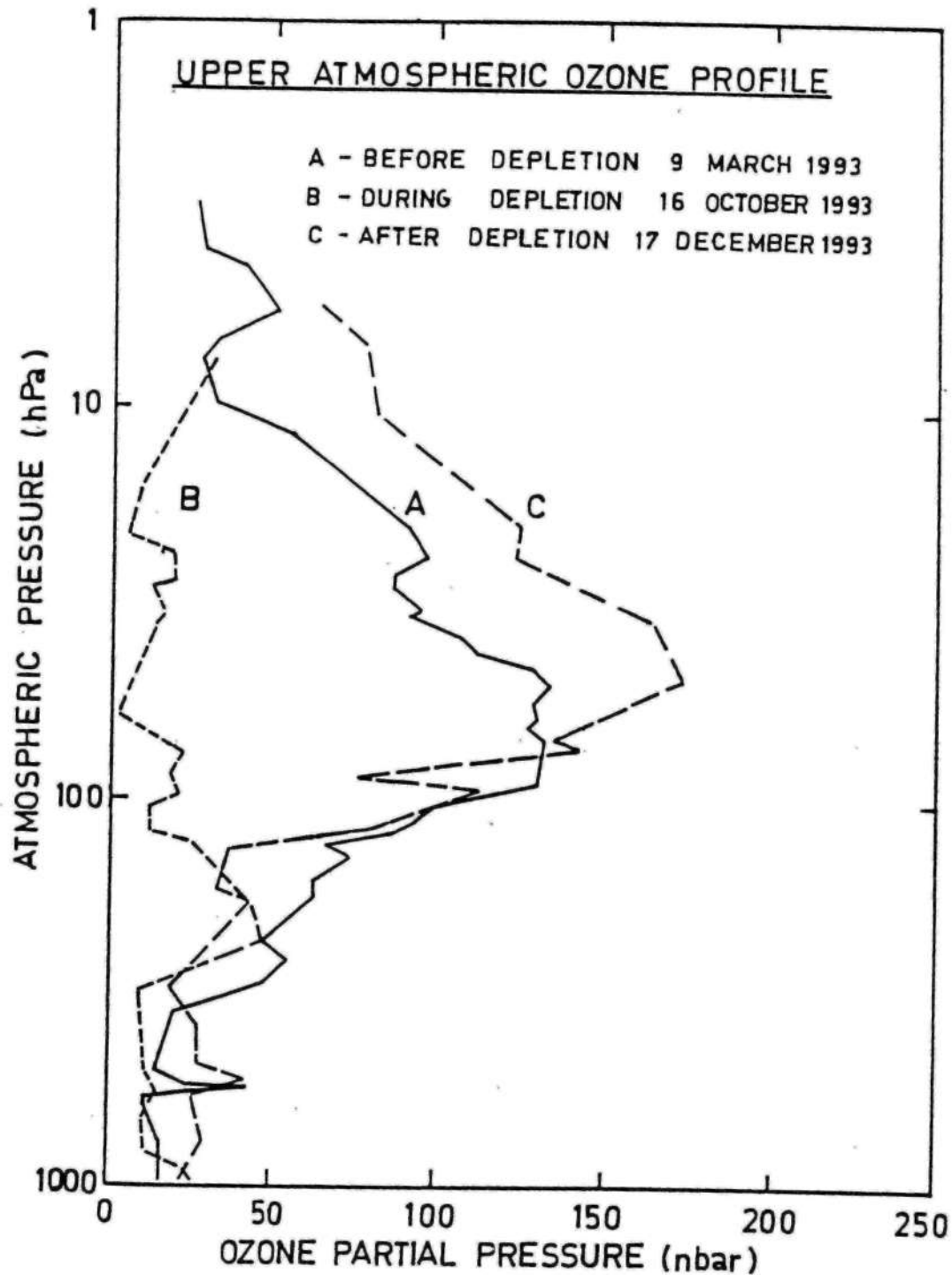


Fig. 4

Case Study of a Blizzard (July-August 1993)

The strongest blizzard occurred at Maitri from 1705 UTC of 31 July 1993 to 1900 UTC of 1 August. Though the duration was not long compared to many other blizzards, this one had typical characteristics of an ideal blizzard as discussed below:

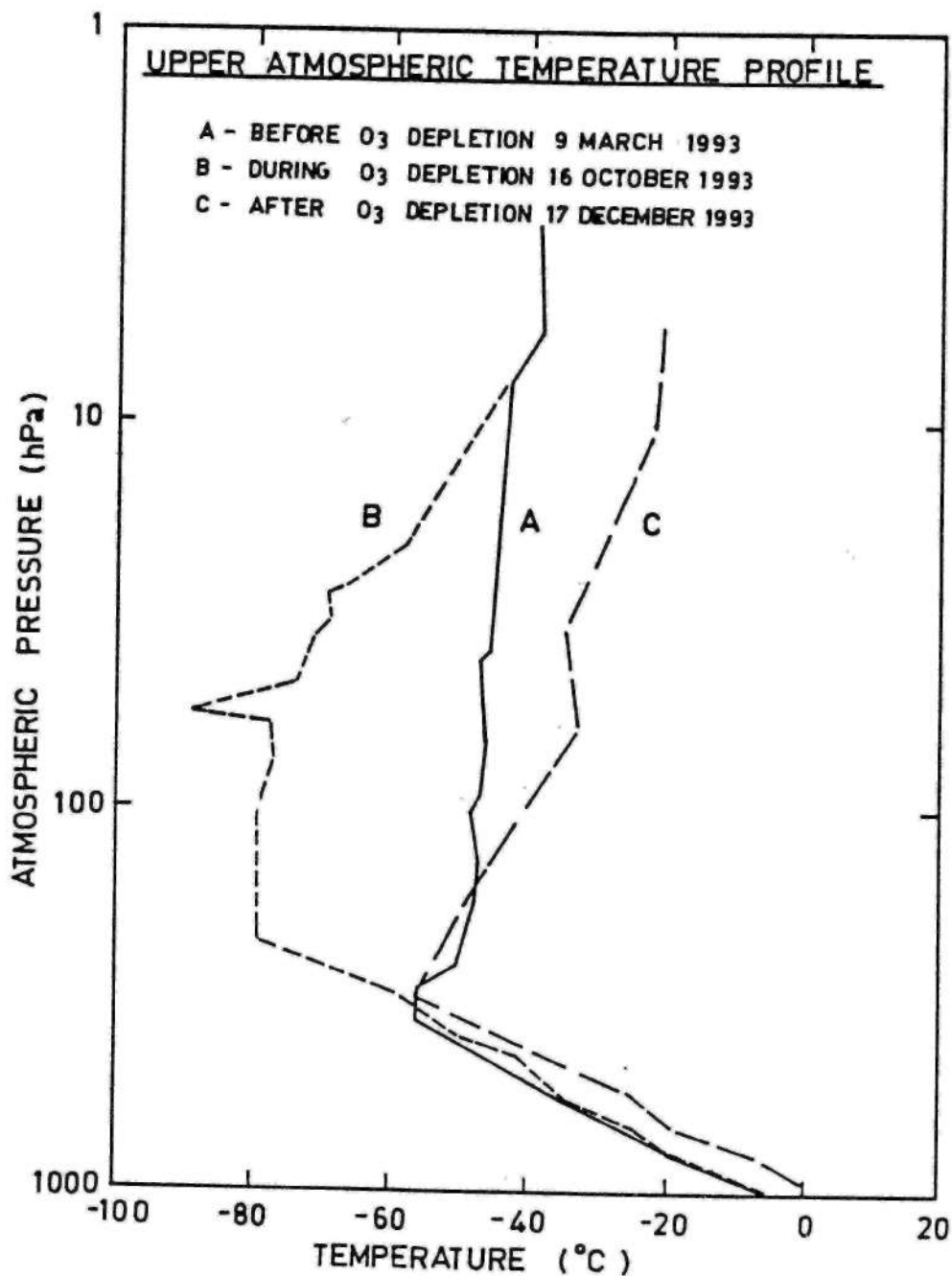


Fig. 5

- (i) Pressure field: Mean sea level pressure fell from 991.1 hPa at 0000 UTC of 31 July to 957.3 hPa at 1200 UTC of 1 August — a fall of 33.8 hPa in 36 hours. It again gradually rose to 977.0 hPa by 0900 UTC of 2 August — a rise of 19.7 hPa in 21 hours.
- (ii) Temperature field: Temperature rose gradually from -22.4°C at 0000 UTC of 31 August to -10.0°C at 0600 UTC of 1 August — a rise of 12.4°C in 30 hours. Again it fell slowly to -12.5°C by 2100 UTC of 1 August. Temperature remained steady around -11°C for few days due

to cloud cover, keeping the heat transported in by the blizzard trapped in the atmosphere over Maitri.

- (iii) Wind field: Average wind rose rapidly from 20 knots at 12 UTC on 31 July to 66 knots at 06 UTC of 1 August and slowly fell to 30 knots by 12 UTC of 2 August. Maximum gust of 92 knots was recorded on 1 August at 0650 hours.
- (iv) Cloud field: Sky became overcast by 12 UTC of 31 July and remained obscured due to blizzard from 1705 UTC till 18 UTC of 1 August. Later too, it remained overcast throughout 2 August.
- (v) Weather field: Severe blowing snow started at 1705 UTC of 31 July with a sudden rise in wind, and then it continued till 1900 UTC of 1 August. Later it subsided abruptly. During this period, visibility recorded on most occasions was less than 50 m.

Cloud Cover

The tropopause being at a lower height in polar regions, the cloud activity is restricted to a very limited height in the atmosphere over Antarctica. Cumulus developments were noticed in summer on few occasions. Strato cumulus clouds were also seen for quite a good number of days. During most of the winter months, altostratus clouds were seen.

It is difficult for clouds to form *in situ* on glaciers as the two essential requirements of moisture and heat (insolation) are not adequate. If we consider the moisture content in the atmosphere, the absolute humidity values are relatively very low even when it is snowing (during which the relative humidity is supposed to be near 100%). Sometimes during summer when the rocky mountain ranges like Wohlthat Mountain gets heated up, causes the formation of very thin clouds locally. These translucent clouds slowly drift along with the wind and form alto-cumulus lenticularis like clouds. Though they are medium clouds by their structure, they are seen at a height of high clouds.

High clouds were also seen frequently, before and after a low pressure system crossing across the coast towards north. The clouds associated were either cirrus in long streak or cirrostratus accompanied by cirrus, which converge at one or two ends near horizon.

There were a few occasions when stratus was also observed, associated with drifting snow or snowfall. When such stratus clouds were observed, the snowfall either moved with the cloud or intensity increased when the cloud passed over the observer. Stratus clouds were generally seen associated with stratocumulus clouds.

One peculiar feature associated with the Antarctic clouds is the distinct cloudiness for longer periods. When it remains clear sky, it remains so for quite a few days, with an isolated altocumulus or a streak of cirrus. When it becomes overcast, it will remain so, with altostratus on most occasions, with a featureless sheet of gray clouds, in which, openings are rarely seen. However, there were a good number of clear days as well as overcast days throughout the year, irrespective of season.

Annual cloud cover analysis indicates that there are about 30% chances of sky either clear or 1/8 covered with clouds and 45% chances of overcast or 7/8 covered. Other possibilities of cloud cover is about 5% each including obscured sky during blizzards. Cold months (March-August) indicate a rise in the probability of clear and 1/8 cloud cover to 36% and a fall in overcast and 7/8 cloud cover to 39%. This analysis indicates that on most occasions, the sky was either near clear state or near overcast state, both in winter or in summer. During summer months some sort of cumulus, tufted clouds in rags/pieces are seen, causing better distribution of cloud cover during these months.

Global Solar Radiation

First sunset at Maitri occurred on January 26 and annual sunset on May 24. Sun returned on July 19 and remained above horizon from November 22. This is because of the positional consideration on globe. However, it may not be true always because Maitri is lying in a valley. Pyranometer observations were recorded during sunny period to the solar energy inflow at Maitri.

Vertical Distribution of Ozone

47 ozonesonde ascents were taken during 1993 to study the ozone and temperature distribution in the upper atmosphere. India also participated in Regional Ozone Soundings in Antarctica (ROSA-93) Programme from September 1-10, 1993. Under this programme, ozone soundings were taken simultaneously at Neumayer (Germany), Syowa (Japan) and Maitri (India) during this period. Four successful ozone ascents were taken at Maitri, eight at Syowa and four at Neumayer. The ozone profiles at Maitri are in reasonable agreement with profiles at Syowa and Neumayer. Some important profiles values are presented in Table 4. Fig. 4 shows how the partial pressure of ozone in the upper atmosphere varied before, during and after the ozone depletion period.

The graph (Fig. 4) indicates in detail, the extent of depletion and subsequent restoration of stratospheric ozone over Maitri in less than 3 months during spring and early summer respectively. The temperature profiles presented indicate that the phenomenon of stratospheric ozone is strongly related to

stratospheric temperature. It appears that colder the winter stratosphere, greater is the depletion of ozone in the ensuing spring.

Ozonesonde data recorded indicate a gradual fall in the maximum partial pressure of ozone from the middle of September to middle of October and a gradual rise thereafter. This is a clear indication of the formation of ozone-hole in the Antarctic spring months. The millibaric pressure level at which the highest partial pressure of ozone occurs also remained steady around 70 mb until the start of depletion after which the maxima shifted either lower around 250 mb or higher around 025 mb pressure levels. This is because of severe depletion at the normal level around 70 mb.

The ozone concentration in the troposphere did not show any significant variation due to severe mixing on account of strong winds in this layer, where intense meteorological phenomena occurs.

Local Forecasting

The IMD team continuously monitored the weather systems for local forecasts. For achieving this, a record number of satellite pictures were received throughout the year, in addition to weather charts on the facsimile recorder. Weather is well predictable at Antarctica with proper monitoring of the systems moving around continuously.

Proper briefing was provided to 13 convoys and teams on scientific activities like ice-core drilling, geological sampling and biological sampling at the ice shelf during the year.

Snowfall

Snowfall is seen all round the year though larger number of days experienced snowfall in deep winters and spring months over autumn and summer months. Snowfall is generally associated with blizzards which causes blowing and drifting snows. Visibilities of less than 10 metres were noticed in intense snowfall associated with winds. When the wind blows, the snow on land is being blown off and the net deposit remains negligibly small. There were, however, some occasions, when there was snowfall with light or no wind and snow accumulated as much as 20 cm in a day.

Aurora

Regular auroral observations were taken to augment the geomagnetic observations. Maximum number of auroras are seen in April. On most occasions, auroras occurred around 2300 hrs and lasted upto 0300 hrs next day.

However, there were occasions when auroral displays lasted for as little as 30 minutes to as long as 10 hrs.

Acknowledgements

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